

Module 7 - Datatypes and Their Types

1. What is a Datatype?

Datatype is a type of value that defines what kind of data a variable can store.

2. What are Primitive Datatypes?

Primitive Datatypes are basic data types that store a single value, such as Number, String, Boolean, Undefined, and Null.

3. What is a Number?

Number is a datatype used to represent numeric values, including integers and decimal numbers.

Example Code:

```
// Number  
var num = 120;  
var num = 1.5;  
console.log(num); // 1.5
```

4. What is a String?

String is a datatype used to represent text or a sequence of characters, enclosed in single quotes (' '), double quotes (" "), or backticks (` `).

Example Code:

```
// String  
let userName = "Javascript is a Scripting Language";  
userName = 'Javascript is a Single threaded Language';  
userName = `ECMA Script`;  
console.log(userName); // ECMA Script
```

5. What is a Boolean?

Boolean is a datatype used to represent two possible values: true or false.

Example Code:

```
// Boolean  
let condition = true;  
condition = false;  
console.log(condition); // false
```

6. What is Undefined?

Undefined is a datatype that represents a variable that has been declared but not assigned a value.

Example Code:

```
// Undefined
// let noValue = undefined;
let noValue;
console.log(noValue); // undefined
```

7. What is Null?

Null is a datatype used to represent an intentional absence of a value.

Example Code:

```
// Null
let emptyValue = null;
console.log(emptyValue); // null
```

8. What are Non-Primitive Datatypes?

Non-Primitive Datatypes (Reference types) are used to store collections of values or complex data, such as arrays and objects.

9. What is an Array?

An array is a data structure used to store multiple values in a single variable. Each value is stored at a specific index, starting from index 0.

Example Code:

```
// Array
// Length          1          2          3          4
let multipleUsers = ["React Js", "Javascript", "Facebook", "Instagram"];
// Index           0          1          2          3
console.log(multipleUsers);
console.log(multipleUsers[0], multipleUsers[1], multipleUsers[3]);
console.log(multipleUsers.length);
console.log(multipleUsers.length-1);
console.log(multipleUsers[multipleUsers.length-1]);
```

10. What is an Object?

An Object is used to store related information in key-value pairs, where each key represents a property and its value contains the corresponding information.

Example Code:

```
// Object (Key value pair)
let vehicle = {
  vehicleType : "Four Wheeler",
  brand : "Hyundai",
  price : 100000,
  fuelType : "Petrol"
};
console.log(vehicle);
console.log(vehicle.price);
console.log(vehicle.brand);
```

11. What is a Single-Line Comment?

A Single-Line Comment is used to write a comment on one line of code and is written using `//`.

Example Code:

```
// Single line comment
```

12. What is a Multi-Line Comment?

A Multi-Line Comment is used to write comments across multiple lines of code and is written using `/* */`.

Example Code:

```
/*
  Multi line comment
  Javascript
  React Js
*/
```